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Preface

- Part **I** contains the questions for the exam
 - These are divided into four blocks: A, B, C, and D.
 - Block A is compulsory and must be answered by all candidates.
 - You must then select ONE block out of B, C, and D.
 - Should you change your mind during the exam, indicate clearly which block you wish to have marked.
 - You can achieve a maximum of 120 points.
 - Information on the maximum number of points obtainable is outlined at the beginning of each block
- Part **II** contains the case study to which all questions in Block A of Part **I** refer
- Part **III** contains the relevant statistical tables
 - Normal curve tail probabilities are provided in Table **3**, page **10**.
 - t-distribution critical values are provided in Table **4**, page **11**.
- Part **IV** contains the formula collection and notation of the module
 - If you use notation other than in the formula collection (page **12**) or notation table (page **13**), give an explanation.
- Allowed time: 2 hours.
- A university-approved calculator is allowed.

Part I: Exam Questions

1 | Block A – Compulsory

80 points, points for each question according to level

All questions in this section relate to the case study in Part II.

Level 1 (each question 4 points)

1. Subsection 7.1: what does `header=TRUE` achieve?
2. Figures 1 and 2 relate to the variable `crime_rate` in the same data frame, `simd2`. How do these Figures differ from one another?
3. Verbalise the code in Subsection 7.2.

Level 2 (each question 6 points)

4. Relate the information provided in Subsection 7.3 with the shape of the distribution depicted in Figure 1.
5. Interpret the point on the cumulative distribution function indicated by the dotted lines in Figure 2.
6. Interpret the output of Subsection 7.5
7. We suspect that alcohol-related admissions to hospital (ALCOHOL) influences the standardised mortality ratio (SMR). Assume you wished to establish causality in your analysis. Name and explain two steps to achieve this.
8. Rewrite the recoding of `drive_cat` in Subsection 7.4 with the option `right = TRUE`.

Level 3 (each question 8 points)

9. Subsection 7.6: What is the sample mean of `simd$drive_primary`?
10. Consider Figure 3. Evaluate this graph, naming, and explaining one problem with this graph, and address the problem by altering the following code:

```
ggplot(simd, aes(x=drive_cat, fill=rank_cat)) +  
  geom_bar() +  
  ylab('Frequency')
```

2 | Block B – True / False Statements

40 points, 5 points for each statement

State whether each of the following statements is true or false. Explain why the statement is true or false.

1. If the mean of a variable is larger than its median, the distribution of this variable is positively skewed.
2. With a smaller effect size of interest, a higher sample size is required to achieve the same level of statistical power.
3. The p-value and the α -level are the same thing.

3 | Block C – Calculations

40 points, points for each question in parentheses

For the following questions you are required to carry out calculations. Show the calculations in detail and explain the steps you are taking.

1. Using the data presented in Table 1, calculate:
 - a. the mean of x . **(2)**
 - b. the standard deviation of x . **(2)**
 - c. the 65th percentile of x . **(5)**
2. A political scientist collects data from a random sample of 40 local councillors to estimate the average number of community meetings they attend each month. The sample has a mean of 12 meetings, with a standard deviation of 3.5 meetings. What is the upper limit of the 99% confidence interval for the true average number of community meetings attended monthly by all local councillors? Round your answer to three decimal places. **(6)**

Obs.	x
1	300
2	460
3	180
4	245
5	259

Table 1: Data Set - Block C

4 | Block D – Explanations

40 points, 8 points for each method / concept

Explain each of the following methods or concepts in your own words, and illustrate your explanation with a worked example of your choice.

1. Central Limit Theorem
2. Statistical Power

Part II: Case Study

5 | Preface

This case study resembles a shorter version of one you will receive for the exam.

Today you will be working with the “Scottish Index of Multiple Deprivation”. The data are taken from Scottish Government (2020).

The Scottish Index of Multiple Deprivation is a relative measure of deprivation across 6,976 small areas (called data zones). If an area is identified as ‘deprived’, this can relate to people having a low income but it can also mean fewer resources or opportunities. SIMD looks at the extent to which an area is deprived across seven domains: income, employment, education, health, access to services, crime and housing. It ranks data zones from most deprived (ranked 1) to least deprived (ranked 6,976).

SIMD is the Scottish Government’s standard approach to identify areas of multiple deprivation in Scotland. It can help improve understanding about the outcomes and circumstances of people living in the most deprived areas in Scotland. It can also allow effective targeting of policies and funding where the aim is to wholly or partly tackle or take account of area concentrations of multiple deprivation.

Data zones in rural areas tend to cover a large land area and reflect a more mixed picture of people experiencing different levels of deprivation. This means that SIMD is less helpful at identifying the smaller pockets of deprivation found in more rural areas, compared to the larger pockets found in urban areas. SIMD domain indicators can still be useful in rural areas if analysed separately from urban data zones or combined with other data.

6 | Codebook

Variable	Type	Description
Data_Zone	Code	2011 Data Zone
Intermediate_Zone	Name	2011 Intermediate Zone name
Council_area	Name	Council area name
ALCOHOL	Standardised ratio	Hospital stays related to alcohol use: standardised ratio
drive_primary	Time (minutes)	Average drive time to a primary school in minutes
crime_rate	Rate per 10,000 population	Recorded crimes of violence, sexual offences, domestic housebreaking, vandalism, drugs offences, and common assault per 10,000 people
sim_rank	Rank	Rank of data zones from most deprived (ranked 1) to least deprived (ranked 6,976)
SMR	Standardised ratio	Standardised mortality ratio

Table 2: Codebook for simd_exam Data Set

7 | Output

7.1 Import Data

```
simd <- read.csv('simd.csv', header=TRUE)
```

7.2 Filter

```
simd2 <- filter(simd, crime_rate<400)
```

7.3 Summary

```
summary(simd2$crime_rate)
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   10.83   94.43  161.48  177.65  250.05  399.95
```

7.4 Recoding

```
simd2 %>%
  mutate(drive_cat = cut(drive_primary,
                        breaks = c(0, 2.79, 3.572, 30),
                        labels = c("Short", "Average", "Long"),
                        right = FALSE,
                        include.lowest = TRUE)) -> simd2

simd2 %>%
  mutate(rank_cat = cut(sim_rank,
                        breaks = c(0, 3488, 5232, 6977),
                        labels = c("Most Deprived", "Deprived", "Least Deprived"),
                        right = FALSE,
                        include.lowest = TRUE)) -> simd2
```

7.5 Inference 1

```
t.test(simd2$crime_rate,
      mu = 200,
      alternative = "two.sided")
##
## One Sample t-test
##
## data:  simd2$crime_rate
## t = -16.103, df = 4969, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 200
## 95 percent confidence interval:
##  174.9249 180.3677
## sample estimates:
## mean of x
## 177.6463
```

7.6 Inference 2

```
sd(simd_sample$drive_primary)
## [1] 1.34018

t.test(simd_sample$drive_primary, alternative = "less", mu = 5)
##
## One Sample t-test
##
## data:  simd_sample$drive_primary
## t = -97.62, df = 4499, p-value < 2.2e-16
## alternative hypothesis: true mean is less than 5
```

8 | Graphs

8.1 Graph 1

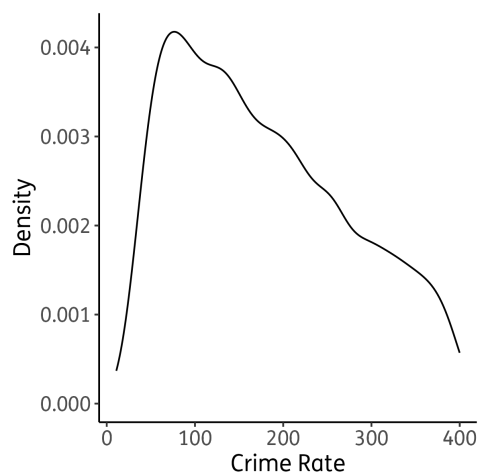


Figure 1: Distribution

8.2 Graph 2

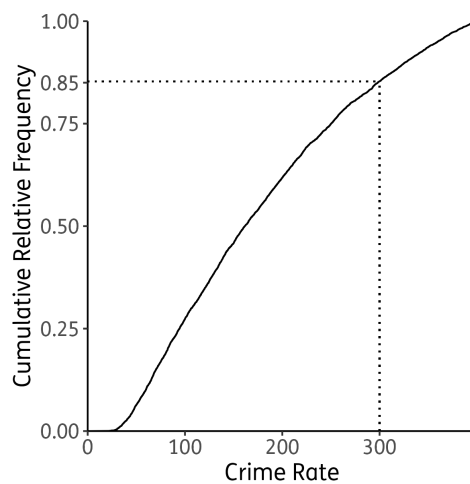


Figure 2: Cumulative Distribution Function

8.3 Graph 3

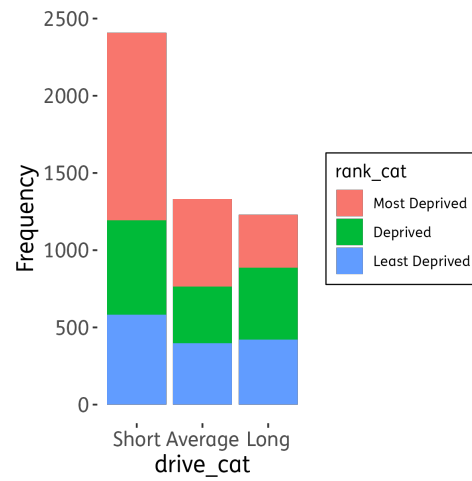
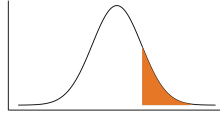


Figure 3: Bad Graph

Part III: Statistical Tables

9 | Normal-Distribution



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002

Table 3: Right-Tail Probabilities under the Normal Distribution

10 | t-Distribution

df	Confidence Level									
	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	One-Tail Probability									
	$t_{0.25}$	$t_{0.20}$	$t_{0.15}$	$t_{0.10}$	$t_{0.05}$	$t_{0.025}$	$t_{0.01}$	$t_{0.005}$	$t_{0.001}$	$t_{0.0005}$
1	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.3	636.6
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.33	31.60
3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.22	12.92
4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291

Table 4: Probabilities under the t-Distribution

Part IV: Formulae and Notation

11 | Formulae

Statistic	Formula
Confidence Interval	$Pr(\bar{y} - t_{\alpha/2} \cdot se \leq \mu \leq \bar{y} + t_{\alpha/2} \cdot se) = 1 - \alpha$
Deviation	$d = y_i - \bar{y}$
Mean	$\bar{y} = \frac{\sum y_i}{n}$ $\mu = \sum y P(y) = E[y]$
Position of p th percentile	$P = (n + 1) \cdot \frac{p}{100}$
Range	$y_{range} = y_{max} - y_{min}$
Standard Deviation	$s = \sqrt{\frac{\sum (y_i - \bar{y})^2}{n - 1}}$
Standard Error	$\sigma_{\bar{y}} = \frac{\sigma}{\sqrt{n}}$ $se = \frac{s}{\sqrt{n}}$
t-test	$t = \frac{\bar{y} - \mu_0}{se}$
Variance	$s^2 = \frac{\sum (y_i - \bar{y})^2}{n - 1}$
z-score	$z = \frac{y - \mu}{\sigma}$

Table 5: Formulae for PO11Q

Symbol	Explanation
d	Deviation
n	Sample Size
s	Standard Deviation
s^2	Variance
$\bar{y}$	Mean
y_i	Observation i
f	(Absolute) Frequency
cf	Cumulative (Absolute) Frequency
rf	Relative Frequency
crf	Cumulative Relative Frequency
$E[x]$	The expected value of x
μ	Mean of the Population
se	Standard Error (with s of sample)
σ	Standard Deviation of the Population
$\sigma_{\bar{y}}$	Standard Error (with σ of population)
t	t-value
z	z-value

Table 6: Notation

Part V: References

Scottish Government. (2020). Scottish Index of Multiple Deprivation 2020 (SIMD 2020). <https://www.gov.scot/collections/scottish-index-of-multiple-deprivation-2020/>